



Minimum Retained Engineering Squad

Staffing brief for the Mubarak AI production scaling
and support engagement

Prepared by Mubarak AI team
September 2026

Minimum team

We are looking for a small, senior team that can take practical responsibility for Mubarak AI. The vendor should propose a development team who will work consistently on the platform, learn the existing architecture, and remain accountable through delivery and support.

The minimum allocation is 3.5 full-time equivalents. A 1.0 FTP role represents one person's full working capacity. A 0.5 FTP role represents roughly half of one person's working capacity.

Allocation

Role	Allocation	Primary responsibility
Lead Full-Stack TypeScript Engineer	1.0 FTP	Application delivery, APIs, integrations, code quality, and day-to-day technical leadership
Senior PostgreSQL and Supabase Engineer	1.0 FTP	Database design, Row Level Security, data integrity, migrations, and query performance
AI and Retrieval Engineer	1.0 FTP	Grounded tutoring, document retrieval, AI quality, model safety, latency, and cost
DevOps and SRE Specialist	0.5 FTP	Deployment, monitoring, reliability, capacity testing, recovery, and incident response

How the team should operate

These roles should operate as one delivery team. The Lead Full-Stack TypeScript Engineer should coordinate the daily technical work, while each specialist remains responsible for decisions and evidence in their area. The vendor should also explain how the team will cover absences, urgent incidents, and knowledge transfer.

The current phase is capped at 300 students through December 2026. The next phase is expected to support up to 5,000 students from January through December 2027. These figures describe the student population; they do not assume that every student will be online at the same time.

Lead Full Stack TypeScript Engineer

This person will be the vendor's day-to-day technical lead and our main engineering contact. They should be comfortable working across the user interface, server-side services, APIs, and third-party integrations rather than focusing on only one layer of the application.

Role in the engagement

The engineer will coordinate technical delivery across the squad, turn agreed requirements into workable tasks, review implementation decisions, and make sure changes fit the existing Mubarak AI architecture.

Core responsibilities

- Maintain and improve the Next.js 16, React 19, and TypeScript application.
- Develop the student, parent, school administrator, and platform operator experiences.
- Build and maintain API endpoints, business services, authentication flows, and external integrations.
- Lead code reviews and keep application changes consistent, tested, and documented.
- Support Arabic and English journeys, right-to-left layouts, accessibility, and responsive behaviour.
- Coordinate releases with the database, AI, and operations specialists.
- Investigate production defects and remain involved until the team has verified the resolution.

Experience we expect

We expect strong, recent experience with TypeScript, the Next.js App Router, React, server-side APIs, authentication, automated testing, and production web applications. Experience with subscription payments, Arabic interfaces, and technical leadership is important for this engagement.

Expected outputs

- Production-ready application and API changes.
- Clear technical plans, reviewed pull requests, and release notes.
- Automated tests for important user journeys and integrations.
- Up-to-date application documentation and handover notes.

Senior PostgreSQL and Supabase Engineer

This engineer will own the data layer and the rules that decide who can see or change each record. The role is especially important because Mubarak AI separates information across students, families, schools, and platform operators.

Role in the engagement

The engineer should treat Row Level Security as a core product safeguard. They will work closely with the application engineer so that database policies, server-side permissions, and user roles enforce the same access model.

Core responsibilities

- Review the existing PostgreSQL schema, Supabase configuration, migrations, indexes, and privileged functions.
- Verify Row Level Security and storage policies for students, parents, schools, and platform operators.
- Create automated tests that attempt cross-user, cross-family, and cross-school access.
- Design safe, repeatable migrations and data-repair procedures.
- Improve slow queries and tune pgvector searches as usage grows.
- Support Supabase Authentication, Realtime, and Storage where they affect permissions or data integrity.
- Test backup and restoration procedures and document the recovery steps.

Experience we expect

We expect senior experience with PostgreSQL, Supabase, Row Level Security, multi-tenant applications, query tuning, migrations, and backup and recovery. The engineer should be able to explain security decisions clearly and demonstrate how the policies were tested.

Expected outputs

- A reviewed and documented database design.
- Verified RLS and storage policies with repeatable test evidence.
- Safe migration scripts and rollback instructions.
- Performance findings, query improvements, and recovery evidence.

AI and Retrieval Engineer

This engineer will be responsible for the quality and safety of the tutoring experience. Their work begins before the model generates an answer: curriculum documents must be processed correctly, the right material must be retrieved, and the answer must stay grounded in that material.

Role in the engagement

The role covers the full retrieval-augmented generation process, including document ingestion, OCR, chunking, embeddings, vector search, reranking, model selection, evaluation, and protection against unsafe or misleading responses.

Core responsibilities

- Maintain the curriculum and textbook ingestion process, including OCR and document structure.
- Improve chunking, embeddings, pgvector retrieval, and reranking so the tutor receives the most relevant source material.
- Build evaluations that measure grounding, answer quality, safety, latency, and cost.
- Test prompt injection, malicious documents, data-exfiltration attempts, and unsafe response paths.
- Maintain the multi-model registry, provider fallbacks, and model-selection rules.
- Improve voice-session reliability and response time where voice tutoring is used.
- Monitor provider usage and recommend practical cost controls without weakening answer quality.

Experience we expect

We expect hands-on experience operating RAG systems in production, not only building demonstrations. The engineer should understand embeddings, vector databases, semantic retrieval, reranking, AI evaluations, prompt-injection controls, model APIs, and the trade-offs between quality, speed, and cost.

Expected outputs

- A documented and repeatable content-ingestion process.
- Retrieval and answer-quality evaluations with agreed pass criteria.
- Security test results and remediation evidence for AI-specific risks.
- Model, latency, and cost reports with clear operating recommendations.

DevOps and SRE Specialist

This specialist will keep the platform deployable, observable, and recoverable. The role is allocated at 0.5 FTE for normal operations, although the vendor may need to propose additional effort during transition, performance testing, major releases, or serious production incidents.

Role in the engagement

The specialist will connect software delivery with production operations. They should give the team reliable deployment pipelines, useful monitoring, tested recovery procedures, and evidence that the platform can meet the two student-capacity milestones.

Core responsibilities

- Manage development, staging, and production configuration across Vercel, Supabase, and connected services.
- Maintain CI and CD pipelines, protected release checks, secrets handling, and rollback procedures.
- Configure dashboards and alerts for availability, latency, errors, queues, database health, and provider failures.
- Design and run staged load, stress, and soak tests for the 300-student and 5,000-student phases.
- Maintain operational runbooks, incident escalation, and communication procedures.
- Test backup restoration and document recovery time and recovery point expectations.
- Track hosting and third-party usage costs and flag capacity or quota risks before they affect service.

Experience we expect

We expect practical experience with Vercel, Supabase operations, GitHub Actions, monitoring, load testing, incident response, backup and recovery, and cloud cost control. Familiarity with Upstash, Inngest, Sentry, and external AI providers is relevant to the current architecture.

Expected outputs

- Reliable deployment and rollback pipelines.
- Monitoring dashboards, alerts, and incident runbooks.
- Reproducible capacity tests and documented results.
- Backup, restoration, recovery, and monthly cost evidence.

Vendor response requirements

Please identify the person proposed for each role. We want to understand who will do the work, how much time they can commit, and whether the same team will remain available after the initial transition.

Information required for each person

- Working location and normal working hours.
- Earliest available start date and confirmed allocation to Mubarak AI.
- Examples of relevant work, with client-confidential information removed where necessary.
- Responsibilities within the proposed team and the person who will provide backup during absence.

Allocation and continuity

If one person is proposed for more than one role, show how their time will be divided and do not count the same hours twice. Explain how the vendor will manage leave, staff replacement, handover, and loss of a key team member. Any additional quality assurance, security, Arabic-language, or delivery-management role should be identified and priced separately.

Support availability

The core support window is Sunday through Thursday, from 07:00 to 22:00 Arabia Standard Time. The proposal should explain how the named team will cover this window and should price a separate 24-hour P1 on-call option.

What we will review

We will assess the relevance of the proposed people, the credibility of their availability, the clarity of role ownership, and the vendor's plan for continuity.

Regards

Mubarak AI team -September 2026